The overall reaction is,

$2Na^{+}(l) + 2Cl^{-}(l) \rightarrow 2Na(l) + Cl_2(g)$ $\qquad E^{\circ} = -4.07V$

The negative E° value shows that the above reaction is a non spontaneous one. Hence, we have to supply a voltage greater than 4.07V to cause the electrolysis of molten NaCl.

In electrolytic cell, oxidation occurs at the anode and reduction occur at the cathode as in a galvanic cell, but the sign of the electrodes is the reverse i.e., in the electrolytic cell cathode is –ve and anode is +ve.

Faraday's Laws of electrolysis

First Law

The mass of the substance (m) liberated at an electrode during electrolysis is directly proportional to the quantity of charge (Q) passed through the cell.

i.e $m \alpha Q$

We know that the charge is related to the current by the equation $I = \frac{Q}{t} \Rightarrow Q = It$

$\therefore \quad m \alpha It$

(or)

$m = Z\,It$ $\qquad$ ..... (9.33)

Where is Z is known as the electro chemical equivalent of the substance produced of the electrode.

When, I = 1A and t = 1sec, $\qquad$ Q = 1C , in such case the equation (9.32) becomes, (9.33)

$\Rightarrow m = Z$ $\qquad$ .....(9.34)

Thus, the electrochemical equivalent is defined as the amount of substance deposited or liberated at the electrode by a charge of 1 coulomb.

Electro chemical equivalent and molar mass

Consider the following general electrochemical redox reaction

$M^{n+}(aq) + ne^{-} \rightarrow M(s)$

We can infer from the above equation that 'n' moles of electrons are required to precipitate 1 mole of M^{n+} as M(s).

The quantity of charge required to precipitate one mole of M^{n+} = Charge of 'n' moles of electrons

$= nF$

In other words, the mass of substance deposited by one coulomb of charge

$$\text{Electrochemical equivalent of } M^{n+} = \frac{\text{Molarmass of M}}{n\,(96500)}$$

(or)

$$Z = \frac{\text{Equivalent mass}}{96500} \qquad \text{.....(9.35)}$$

XII U9 Electro Chemistry.indd 54 11-01-2021 14:44:19